

IMPLEMENTATION OF CHATGPT USING DJANGO

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CERTIFICATE

This is to certify that the mini project entitled:"**IMPLEMENTATION OF CHATGPT USING DJANGO**"submitted by T. Pranathi , B. Haseena, P. Muneeswari.to the Department of Electronics and Communication Engineering, Annamacharya Institute of Technology and Sciences Tirupati, is a bonafide work carried out by them during VII Semester in partial fulfillment of the requirements for the award of the Bachelor of Technology degree.

Project Guide:

Head of Department:

Principal:

ACKNOWLEDGE

We would like to express our sincere gratitude and heartfelt thanks to all those who have directly or indirectly contributed to the successful completion of our mini project titled "Implementation of ChatGPT Using Django."

First and foremost, we express our deepest gratitude to our respected project guide for providing valuable guidance, encouragement, constructive suggestions, and continuous support throughout the development of this project. Their expertise, patience, and motivation helped us overcome challenges and successfully complete the project.

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1. Abstract

The "Implementation of ChatGPT Using Django" project aims to develop a web-based conversational AI application using the Django framework. The system allows users to interact with an AI chatbot through a user-friendly interface. Django handles the backend operations, user requests, and API integration, while the frontend provides an interactive chat experience. This project demonstrates the integration of artificial intelligence with web development technologies.

2. Introduction

Artificial Intelligence has transformed the way users interact with applications. ChatGPT is an advanced language model capable of understanding and generating human-like responses. This project focuses on implementing a ChatGPT-based chatbot using Django, enabling users to communicate with the AI through a web browser.

3. Problem Statement

Traditional applications provide static information and lack intelligent interaction capabilities. Users require a system that can understand natural language queries and provide meaningful responses. The project aims to build a chatbot that can engage in conversational interactions and improve user experience.

4. Objectives

- To develop a chatbot application using Django.
- To integrate AI-based conversational capabilities.
- To provide a simple and user-friendly chat interface.
- To process user queries and generate intelligent responses.
- To demonstrate practical implementation of AI in web applications.

5. Methodology

1. Install and configure Django environment.
2. Create a Django project and application.
3. Design the frontend interface using HTML, CSS, and JavaScript.
4. Integrate ChatGPT API with Django backend.
5. Accept user input through the chat interface.
6. Send requests to the AI model.

7. Receive and display generated responses.
8. Test the application and optimize performance.

6. Development of ChatGPT

ChatGPT is a large language model developed by OpenAI. It is based on Transformer architecture and is trained using huge amounts of text data.

ChatGPT can:

- Answer questions.
- Generate content.
- Explain concepts.
- Assist in programming.
- Translate languages.
- Support educational activities.

The introduction of ChatGPT significantly improved conversational AI systems because of its ability to understand context and provide human-like responses.

7. System Requirements

7.1 Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB or higher
- Storage: 10 GB free space

7.2 Software Requirements

- Python 3.x
- Django Framework
- Visual Studio Code
- Web Browser (Chrome, Edge, Firefox)
- OpenAI API Key (Optional)

8. Implementation

The project is implemented using Django as the backend framework. Users enter messages through a web interface. Django receives the request and communicates with the ChatGPT API. The generated response is processed and displayed on the webpage in real time.

8.1 System Architecture

User → Web Interface → Django Backend → ChatGPT API → Response → User Interface

9.Source Code

Views.py

```
from django.shortcuts import render

from django.http import JsonResponse

def home(request):

    return render(request, 'index.html')

def chat(request):

    if request.method == "POST":

        user_message = request.POST.get("message")

        # Sample response

        bot_response = f"You said: {user_message}"

        return JsonResponse({"response": bot_response})
```

urls.py

```
from django.urls import path

from . import views

urlpatterns = [

    path("", views.home, name='home'),

    path('chat/', views.chat, name='chat'),

]
```

index.html

```
<!DOCTYPE html>
```

```

<html>
<head>
  <title>ChatGPT Using Django</title>
</head>
<body>
  <h2>ChatGPT Using Django</h2>
  <input type="text" id="message">
  <button onclick="sendMessage()">Send</button>
  <div id="response"></div>
  <script>
    function sendMessage() {
let message = document.getElementById("message").value;
      fetch('/chat/', {
        method: 'POST',
        headers: {
          'Content-Type':
            'application/x-www-form-urlencoded',
          'X-CSRFToken': '{{ csrf_token }}'
        },
        body: `message=${message}`
      })
      .then(response => response.json())
      .then(data => {
        document.getElementById("response").innerHTML =
          data.response;
      });
    }
  </script>

```

</body>

</html>

10. Results

- Successfully created a web-based chatbot application.
- Users can enter questions and receive responses.
- Django efficiently handles backend operations.
- The system provides an interactive conversational experience.

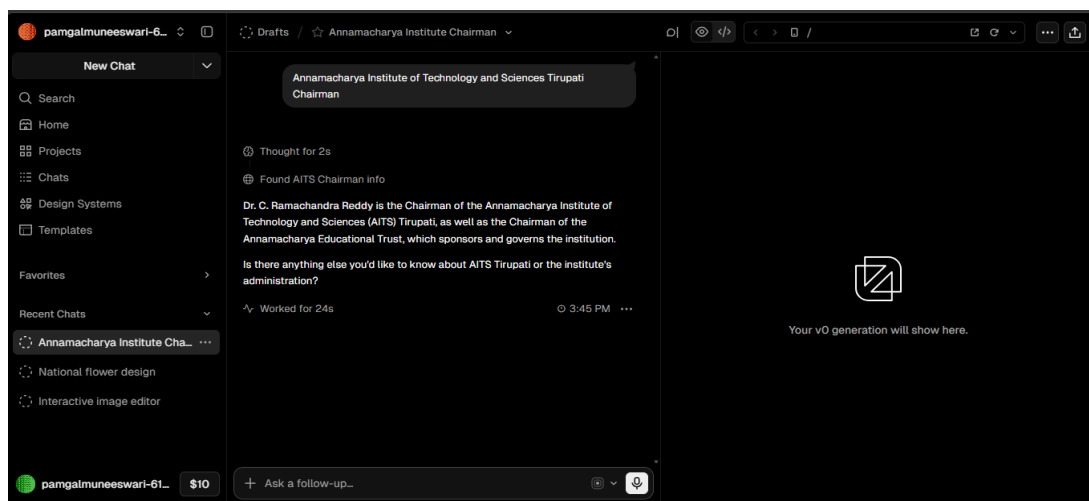


Fig:1 Implementation of chatgpt

11. Advantages

- User-friendly interface.
- Fast response generation.
- Easy to maintain and extend.
- Demonstrates AI integration in web applications.

12.Future Enhancements

- User authentication and chat history.

- Voice input and output support.
- Multi-language support.
- Database storage of conversations.
- Advanced AI features and personalization.

Conclusion

The "Implementation of ChatGPT Using Django" project successfully demonstrates the integration of conversational AI with a web application. Using Django's robust framework and ChatGPT's language capabilities, the system provides an interactive platform for users to communicate with an AI assistant. The project highlights the practical application of artificial intelligence in modern web development and can be further enhanced with advanced features in the future.